SQL Functions and Triggers - Beginner Study Guide

1. What is a Function?

A Function is a database object that accepts input (optional), performs a calculation or logic, and returns a value. Syntax:

```
CREATE FUNCTION dbo.fn_Name(@Param INT)
RETURNS INT
AS
BEGIN
  RETURN @Param * 2
END;
```

Example:
`SELECT dbo.fn_Name(5); -- Returns 10`

2. Types of Functions

Built-in Functions: UPPER(), LOWER(), LEN(), GETDATE(), ROUND(), SUBSTRING().
User Defined Functions (UDF): Functions created by developers.
Types of UDF: Scalar Function (returns one value), Table-Valued Function (returns a table).

3. ALTER and DROP Function

Alter:
`ALTER FUNCTION dbo.fn_Name(@Param INT) RETURNS INT AS BEGIN RETURN @Param*3 END;`

Drop:
`DROP FUNCTION dbo.fn_Name;`

4. What is a Trigger?

A Trigger is a special database object that executes automatically when INSERT, UPDATE, or DELETE occurs on a table. No EXEC statement is required.

5. Trigger Syntax

`CREATE TRIGGER trg_AfterInsert ON Employee AFTER INSERT AS BEGIN PRINT 'Employee inserted' END;`

6. Types of Triggers

AFTER Trigger – Executes after INSERT, UPDATE, or DELETE.
INSTEAD OF Trigger – Executes instead of the requested DML operation.
DDL Trigger – Fires on CREATE, ALTER, DROP database events.
LOGON Trigger – Fires when a user logs into SQL Server (advanced).

7. Trigger Example

`CREATE TRIGGER trg_DeleteAudit ON Employee AFTER DELETE AS BEGIN INSERT INTO EmployeeAudit SELECT * FROM deleted; END;`
The special **deleted** table contains rows removed by the DELETE statement.

8. Function vs Stored Procedure

Function: Returns a value, can be used in SELECT, ideal for calculations.
Stored Procedure: Performs business logic, may or may not return values, executed with EXEC.

9. Trigger vs Stored Procedure

Trigger: Runs automatically when data changes.
Stored Procedure: Runs only when explicitly executed.

10. Quick Revision

• Function = Input → Processing → Return Value
• Trigger = Automatic response to INSERT/UPDATE/DELETE
• ALTER modifies an object; DROP removes it permanently.
• AFTER triggers are the most common for auditing and logging.